

For the problem below:

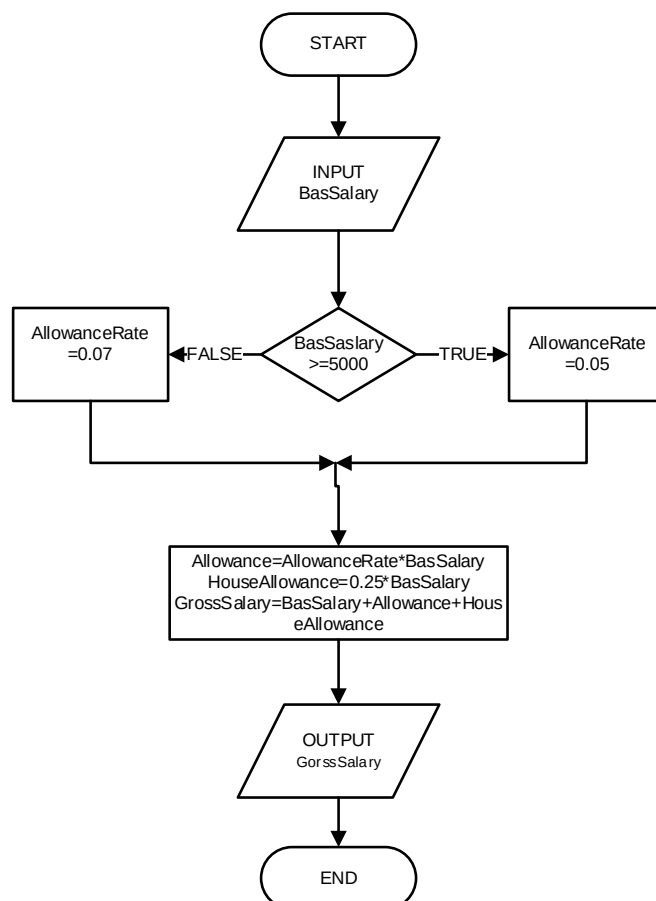
- Analyze the problem by identifying the input, output, formula, and conditions.
- Design an algorithm to solve the problem.
- Prepare, compile, link, and execute the program to solve the problem.
- Test your program using the prepared test data (show the output run).
- Write proper documentation in the program, including the following information to form a banner at the beginning of your program:

```

/*****
* Author: your name and student ID
* Course: the course code only
* Date: of code written
* Brief description: of what problem the program tries
  to solve
*
*****/

```

Write a computer program to calculate the monthly gross salary of a worker at PetroNasa Company. Every month, the basic salary will be entered, and their allowance will be either five percent (5%) or seven percent (7%) of the basic salary, while the house rent allowance will be twenty-five percent (25%) of the basic salary. Your program code must follow the process steps given in the following flowchart. Modify the flowchart to check if an invalid salary was entered (a negative value) and include a user-friendly interface as shown in the provided sample run.



Sample run 1

Please enter Basic Salary (RM): -5000.00 <enter>

Invalid Input. Please rerun the program and enter a valid input.

Sample run 2

Please enter Basic Salary (RM): 3105.00 <enter>

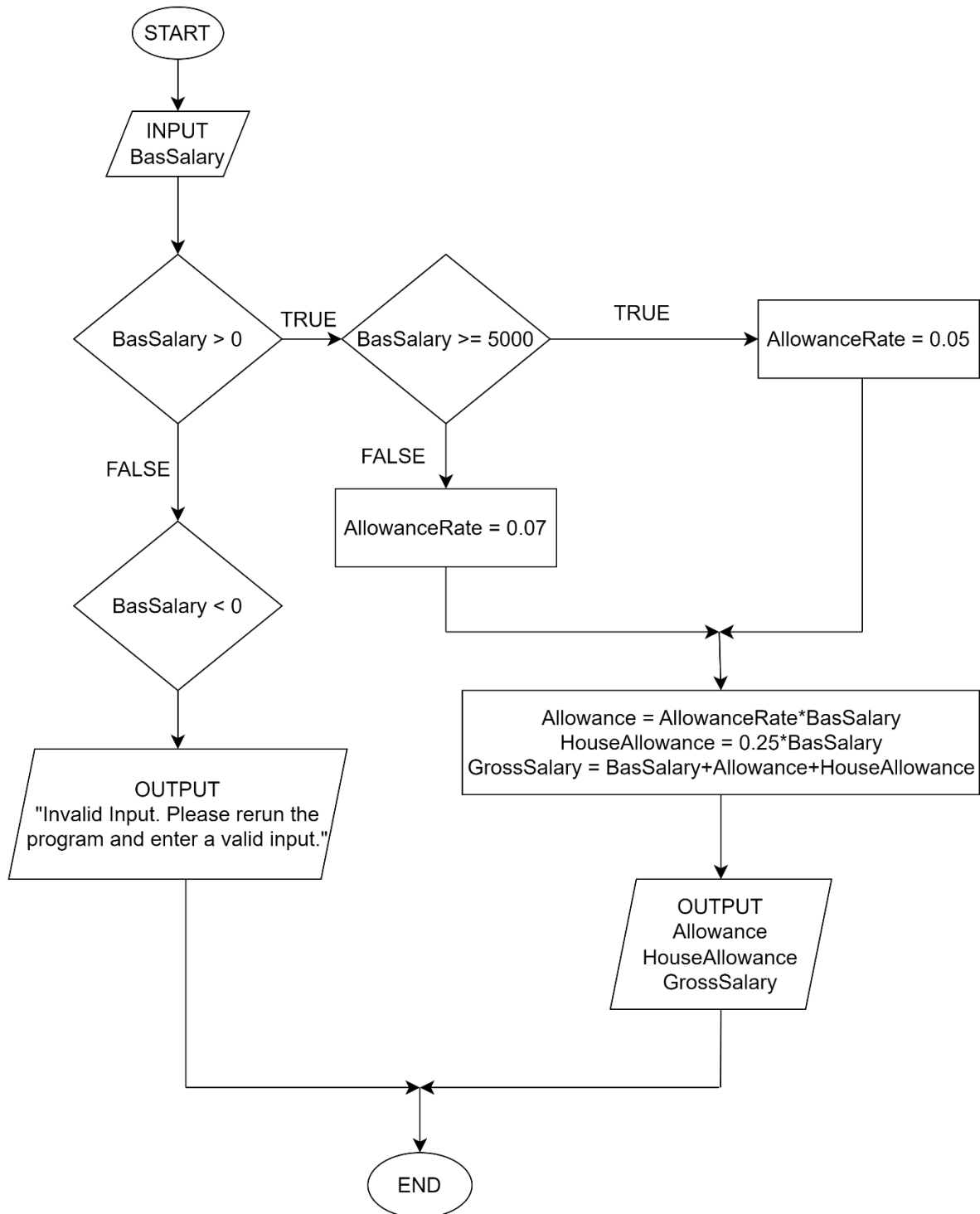
Allowance (RM): 155.25

House Rent allowance (RM): 776.25

Gross Salary (RM):4036.50

Answer:

Flowcharts



Pseudocode

1. Prompt user to enter their BasSalary
2. Read user input (BasSalary)
3. if BasSalary is more than 0 then

 if BasSalary is more than or equal to 5000 then

 Allowance is equal to 0.05

 else

 Allowance is equal to 0.07

 Allowance = AllowanceRate * BasSalary

 HouseAllowance = 0.25 * BasSalary

 GrossSalary = BasSalary + Allowance + HouseAllowance

 Print Allowance

 Print HouseAllowance

 Print GrossSalary

else if BasSalary is less than 0 then print "Invalid Input. Please rerun the program and enter a valid input."

C++ Program

```
/*  
*****
```

Author: Irdina Nadhirah Binti Bahanuddin (23B08I006)

Course: BCSC

Date: 12/10/2023

Brief description: Calculate the monthly gross salary at PetroNasa Company

```
*****  
*****/
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    float BasSalary, GrossSalary, Allowance, AllowanceRate,  
    HouseAllowance; // declare variables
```

```
    cout<<"Please enter Basic Salary (RM): "; // prompt user
```

```
    cin>>BasSalary; // input Basic Salary
```

```
    if (BasSalary > 0)
```

```
    {
```

```
        if (BasSalary >= 5000)
```

```
        {
```

```
            AllowanceRate = 0.05; // declare a Allowance rate to  
0.05
```

```
        }
```

```
        else
```

```
        {
```

```
            AllowanceRate = 0.07; // declare a Allowance rate to  
0.07
```

```

    }

    Allowance = AllowanceRate * BasSalary;           //
calculate Allowance

    HouseAllowance = 0.25 * BasSalary;               //
calculate House allowance

    GrossSalary = BasSalary + Allowance + HouseAllowance; //
calculate Gross salary


    cout<<"\nAllowance (RM): "<<Allowance;           //
output Allowance

    cout<<"\nHouse Rent Allowance (RM): "<<HouseAllowance; //
output House allowance

    cout<<"\nGross Salary (RM): "<<GrossSalary;       //
output Gross salary


}

else if (BasSalary < 0)
{
    cout<<"\nInvalid Input. Please rerun the program and enter a
valid input."; // output Invalid Input
}

return 0;
}

```

Program Testing 1

Please enter Basic Salary (RM): 9574

Allowance (RM): 478.7

House Rent Allowance (RM): 2393.5

Gross Salary (RM): 12446.2

Program Testing 2

Please enter Basic Salary (RM): -4632

Invalid Input. Please rerun the program and enter a valid input.

Program Testing 3

Please enter Basic Salary (RM): 1533

Allowance (RM): 107.31

House Rent Allowance (RM): 383.25

Gross Salary (RM): 2023.56

NAME: IRDINA NADHIRAH BINTI BAHANUDDIN							
MATRIX. NO: 23B08I006							
ATTRIBUTE	CLO	Weighted	Poor 1	Fair 2	Good 3	Excellent 4	Marks
PSEUDOCODE / FLOWCHART	1	2	No working algorithm created.	Incomplete non-working algorithm created with incomplete supporting theory.	incomplete nearly working algorithm created with strong supporting theory.	Complete working algorithm created with strong supporting theory.	/8
SPECIFICATIONS	1	3	The program produces correct results only for some parts.	The program produces correct results but does not display them correctly.	The program works and produces the correct results and displays them correctly. It also meets most of the other specifications.	The program works and meets all of the specifications.	/12
DOCUMENTATION	1	1	The documentation is simply comments embedded in the code and does not help the reader understand the code.	The documentation is simply comments embedded in the code with some simple header comments separating routines.	The documentation consists of embedded comment and some simple header documentation that is somewhat useful in understanding the code.	The documentation is well written and clearly explains what the code is accomplishing and how.	/4
DELIVERY	1	1	The program was delivered within 2 (TWO) weeks of the due date.	The program was delivered within a week of the due date.	The program was delivered within 1 (ONE) days of the due date.	The program was delivered on time.	/4